RAHEE: a digital version of a travel board game

Summary. A web-based, local-multiplayer version of RAHEE, a finished board game in which players journey across a map of India — moving between connected cities, visiting tourist places, collecting themed memories, managing a travel budget, and reaching character destination goals. The team will build a reusable, data-driven RAHEE game engine and local multiplayer interface, in which the complete board and character configuration are external data and the engine contains only the rules and mechanisms needed to play them. The core requirement is that the game is data-driven: the board, cities, routes, tourist places, and destination goals all live in editable data files that the game engine reads at load time and renders over a supplied background map.

editable data file that the game engine reads at load time, rendered over a supplied background map. Changing the data — adding a city, retyping a route, editing a destination goal — changes the game, with no change to the engine code. The minimum version supports complete 2–4-player games on one device. Support for 5–6 players, a visual map editor, and a greedy AI opponent are stretch goals.

Core architecture

RAHEE is implemented as a data-driven game engine. Static game configuration is supplied through structured data files; the engine contains no board-specific knowledge. The engine reads the data at load time, validates it, constructs the game graph and decks, renders the supplied background plate and data-defined elements, and enforces the rules. Changes to cities, routes, tourist places, characters, destination goals, deck composition, or associated assets must therefore be possible without modifying engine code.

Players and modes

- 2–4 human players, local hotseat on one device.
- Private handover screen between turns so each player's cards stay hidden.
- Optional stretch: 5–6 human players.

- Optional stretch: one or more greedy AI players filling empty seats.

The board data file (engine reads this)

- **City / node:** identifier, display name, position in board-image coordinates, and whether it is a tourist place.
- **Route / edge:** the two cities it joins, route type (road / flight / ship), step count, and any flight or ship surcharge.
- **Destination goals:** the destination cities listed on each character, held as data — so editing a character's goals is a data change.
- **Tourist places & assets:** which cities are tourist places, plus references to card, memory-suit, token, and character assets.

Because all of this is data, the board can be changed by editing the file even before any editor UI exists — the visual editor (stretch) is the convenient way to edit it, not the only way.

Screens (minimum set)

- Setup — player names, player count, first-player selection, character selection.
- Handover — hides the previous player's information before showing the next player's hand.
- Board / play screen — background plate with cities, routes, and tokens drawn from the data; current player's hand, budget, memories, destinations, tourist visits, and road-trip cards.
- Turn-action interface — play a card, choose a legal route, pay travel costs, earn memories, visit a tourist place, offer or accept a road trip.
- Scoreboard — current game state and final score breakdown.

Game logic to implement (ruleset provided)

- Movement along connected routes, including step costs and special flight and ship charges — all read from the route data.

- Cards, hidden hands, played road-trip cards, and the different deck composition for different player counts.
- Memories — efficient travel, meeting another traveller, and travelling together on a road trip, across the four suits.
- Tourist visits and per-player one-time marking.
- Road trips — invitation and acceptance, route choice, cost sharing, shared-tourism discount, and intermediate drop-off.
- Aces' three uses: played alone, as a movement booster, or to create an additional memory.
- Character starting cities and destination goals (from data), end-game trigger, final round, budget penalty, and full scoring.
- Overspending is permitted but produces the specified end-game penalty.

Scope

Must have (v1)	**Nice to have / stretch**
• Data-driven engine: board, cities, routes, tourist places, and destination goals all read from an editable data file. • Render cities, routes, and tokens over the supplied background plate from that data. • Editing the data file changes the game with no engine code change. • Full rule enforcement for a legal 2–4 player game. • Local hotseat with a private handover screen. • Complete movement, cards, memories, tourism, road	• Visual map editor: place/move cities, edit routes, edit destination goals, write valid data back. • Swap the background plate through the editor. • Support for 5–6 human players. • Optional greedy AI player (legal, current-turn heuristic moves). • Move / route hints using standard shortest-path routines. • Undo the most recent action.

trips, Aces, destinations, and budget rules. • Automated money, memory, destination, tourist-visit, and scoring records. • Save and resume a game. • End-game trigger and transparent final score breakdown. • Tests for representative rule and scoring situations.	• Simple animations for movement and scoring. • Playtest statistics (turns taken, routes travelled).

Explicitly out of scope

- No online multiplayer or remote networking.
- No geocoding or real-world map positioning — cities are placed in the board image's own coordinates, not by latitude/longitude.
- No automatic extraction of cities or routes from a scanned or drawn map, and no deriving route connections from geography — the connections are game design and are provided by us.
- No game-solving, optimal-play analysis, or novel AI — the optional bot performs only legal, greedy heuristic moves.
- No novel pathfinding algorithm — standard graph and shortest-path routines are sufficient.
- No redesign of the rules, artwork, or character system by the student team.

What we hand over at kickoff

- The complete RAHEE ruleset.
- The background map plate (terrain and decoration, no city markers or route lines) in an agreed image or vector format.
- The place list and — critically — the connection data: which cities join, route types, step counts, and surcharges.

- The character cards: starting cities and destination goals.
- Card, memory-suit, tourist-place, token, and character assets.
- An agreed schema for the board data file, plus sample game states and scoring examples for testing.

Deliverable. A deployable, rule-complete, local-hotseat web version of RAHEE for 2–4 players, whose board — cities, routes, tourist places, and destination goals — is read from an editable data file rendered over the supplied background plate. The game maintains its own state, saves and resumes, and produces a clear final score breakdown. Changing the data file changes the game without rewriting the engine.

Open decisions for the scope meeting. (1) the board-data schema and required fields; (2) format and dimensions of the background plate and other assets; (3) which stretch goal comes first — the visual map editor (reinforces the data-driven design) or the greedy bot.